Formel und Graphen in Anlehnung an
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1 Overview

The content of this document belongs to several different topics and separated documents. The actual collection is therefore only momentarily available and the formatting is reduced to minimum overhead (no title author pages, no TOC). Extensive literature references exist and will be added shortly.

- Theoretical description of all knowledge related to Doppler sensor
  This was started by Florian Exner under the prel. title "Filtertest" and contains already a lot of details about the DRS05 signal processing (all filters).

- Theoretical description of possible new signal processing algorithms
  This part will be a separate document which evaluates the new information and derives the requirements for the development. These final requirements will be handled in JAMA.

- Validation concept for SIL2 Obsolecense Project VEK2410)
  The development needs a concept for validation in several distinct or abstract levels.

  - Correct description of the signal properties and the derived assumptions
  - Test cases where assumptions brake down slowly or completely
  - Unified view on algorithms which are sharing the same concepts
  - Performance of the algorithm according to requirements (abstract level)
  - Correct implementation of the algorithm in a high level language (software)
  - Correct transfer of the implementation to the embedded system (runtime)
  - Correct function of the product (trials)

- Handling of large numbers of formulas is related to a very separate context.
  Actual process is manual, most larger derivations were edited in eqneditor.php and then copied back to align blocks in Latex. For more details see `https://cadabra.science/` release notes or similar programs like KLatex.

2 Contribution to the Validation Concept (best possible algo today)

The validation process should include a comparison with the actual state of the art. A full literature survey and a bench-marking of the existing algorithm is required. If some reference contain already some comparison, these can used as an additional input. The important information is to shown the performance of the product compared to other known algorithm to reduce the risk, that a competitor comes up with an better solution.

3 Contribution to the product strategy (best possible algo in the future)

A literature survey and a bench-marking with upcoming new algorithm should be performed on regular basis to reduce the risk, that a competitor comes up with an better solution. This should not be limited to the same technology, but should be open for other available concepts as long as the requirement could be fulfilled.

4 Contribution to the Validation Concept (all tracks)

The requirement for a sensor to work under all conditions includes also the requirement to work on all types of tracks. Normally it is assumed that the Doppler signal is stochastic for most times. Algorithms assuming this should work fine on ballast track, but if on a part of the track single isolated reflectors exist, then other algorithm will work better. To achieve optimal coverage of the requirement, this situation should be detected and mitigation action should be taken, for example correction of calibration factors or different algorithm. At least a report must be created which compares the theoretical possibilities with the implemented functionality.

5 Contribution to the product strategy (best possible product in the future)

Actual state of the art (2017) is the uage of FPGA signal pre-processing hardware and configurations with 2 or more beam to measure the angles or other additional information, which allows to achieve higher accuracy and reduced effecte due to changes in track conditions. There are several driving forces in separate markets, automotive market, medical system market and computer components (HMI) for gesture detection.

6 Motivation for "Viertel-Perioden" (VP) measurements

VP measurements mean measurements according to the VPH patent, where *"Viertel-Perioden" (VP)* measurements are taken by an simple algorithm or hardware to compare the signals. Measurement data are available or can be generated from recorded Doppler signals by use of simple c-programs. The advantage of the Δt_i and A_i information is the complete information about the signal. Several possibilities exist to derive geometrical information from the measurements. This may already give some understanding without going into details of spectral properties of different track grounds.

$$A(u) = \sum_{i=0}^{i=N} A_r(x_i - u, y_i) \quad (4)$$

$$\mathrm{A}(\alpha) = \left(\sin^2(\alpha) \cdot e^{-\left(\frac{(\alpha-\alpha_0)^2}{2\cdot\beta^2}\right)} \right)^2$$

$A(u) = \sum_{i=0}^{i=N} A_r(x_i - u, y_i)$

$A(\alpha) = \left(\sin^{-2}(\alpha) \cdot e^{-\left(\frac{(\alpha-\alpha_0)^2}{2\cdot\beta^2}\right)} \right)^2$

formula for x,y 2 dimensional

$R(x,H) = \sqrt{x^2 + H^2} => R(x,y,H) = \sqrt{x^2 + y^2 + H^2}$

R from angle in x direction as usual $\quad R = \frac{H}{sin(\alpha)}$

$A(\alpha) = \left(\sin^{-2}(\alpha) \cdot \cos^{-2}(\beta) \cdot e^{-\frac{1}{2}\left(\frac{(\alpha-\alpha_0)^2}{(\Delta\alpha)^2}\right)} \cdot e^{-\frac{1}{2}\left(\frac{(\beta-\beta_0)^2}{(\Delta\beta)^2}\right)} \right)^2$

density based on orig. coordinates $\quad f(x,y) = \sum_{i=0}^{i=N} \delta(x - x_i) \cdot \delta(y - y_i)$density remapped $\quad g(z) = \sqrt{x^2 + y^2} \quad u(z) = \sum$

7 Basic Definitions Part I (section can be moved to the appendix)

7.1 Signal Definition

The output signal from a microwave Doppler transceiver is depending on some reflection inside the antenna characterics. For single moving reflector a pair of voltages I/Q with a defined phase difference will be observed. These voltages are converted to digital values by an ADC. For these digital signals a description as a complex phasor is therefore suitable. The symbols and description follows [?]

$$S(t) = A(t) \cdot e^{i\phi(t)} = A(t) \cdot cos(\phi(t) + i \cdot A(t) \cdot sin(\phi(t) = I(t) + i \cdot Q(t) \tag{5}$$

7.1.1 Special Phase and Amplitude constraints/situations/points

different aspects : deviation from idelaistic model, noise, amplitude error, mis-alignment

- $||I(t)| - |Q(t)|| \leq A_{threshold}$

8 Basic Definitions (section can be moved to the appendix)

8.1 Geometric Setup

Coordinate system is moving, that means it is fixed to the transceiver. It can be assumed that the reflectors are moving towards the transceiver, the opposite case can be added later. If a reflector reach the area, where a the antenna beam illuminated the track ground, then the variables for location and times can be defined as:

$$\text{forward} = \text{reflex moving towards the transmitter} \qquad x(t) = x_o - v \cdot (t - t_o) \tag{6}$$

$$\text{next reflex has other starting point and time offset} \qquad x(t) = x_p - v \cdot (t - t_p) \tag{7}$$

$$\text{for differences within a reflex offsets chancel} \tag{8}$$

$$x(t_2) - x(t_1) = \tag{9}$$

$$x_o - v \cdot (t_2 - t_o) - (x_o - v \cdot (t_1 - t_o)) = -v \cdot (t_2 - t_1) \tag{10}$$

$$\tag{11}$$

The angle α should be the angle between horizontal and reflector.

8.2 Phase Definition

The phase difference between transmitted and received signal is proportional the travelled distance of the wave, that means 2 x the radius vector to the reflector at position x_i (y=0,z=0).

$$k_0 = \frac{2\pi}{\lambda} \tag{12}$$

$$\phi_i = 2 \cdot k_0 \cdot \sqrt{x_i^2 + H^2} \tag{13}$$

$$\tag{14}$$

8.3 Period Duration Measurement

Measurements of the period duration between points with defined phases gives time difference Δt_i, time stamps t_i and amplitudes A_i. For example if measurements are taken at $|I| = |Q|$ points, then the phase difference between measurements is $\pi/2$. Then the number of measurements per 2π is $p = 4$. The index i includes the distance at $x = 0$ and therefore starts with $i_o > 0$ and $\phi_{off} < 2\pi$.

$$\phi_i = \phi_{off} + \frac{2\pi \cdot i}{p} = 2 \cdot k_0 \cdot \sqrt{x_i^2 + H^2} \tag{15}$$

$$\tag{16}$$

For $i = i_o$ we have $x_{i_o} = 0$ and we consume ϕ_{off} by an $i_o \in \mathbb{R}$ and to have i starting with i=0 at x=0, we change the index $i => i_o + i$:

$$i_o = 2 \cdot p \cdot \frac{H}{\lambda} \tag{17}$$

$$x_i = (i + i_o) \cdot \frac{\lambda}{2 \cdot p} \sqrt{1 - (\frac{i_o}{(i + i_o)})^2} \tag{18}$$

$$x_i = (i + i_o) \cdot \frac{\lambda}{2 \cdot p} \sqrt{1 - (\frac{i_o}{i + i_o})^2} = H \sqrt{\frac{(i + i_o)^2}{i_o^2} - 1} = H \sqrt{\frac{i^2 + 2i \cdot i_o}{i_o^2}} \tag{19}$$

The mounting height enters the formula via the values off i_o.

8.4 Distance Differences Δx_i from VP measurements

From x_i we derive the difference of x between measurements, $i > i_o$ for all i. As shown above, the offsets within one reflex cancel out.

$$x(t_2) - x(t_1) = -v \cdot (t_2 - t_1) \tag{20}$$

$$x_i = \cdot \frac{\lambda}{2 \cdot p}(i + i_o)\sqrt{1 - (\frac{i_o}{(i + i_o)})^2} \tag{21}$$

$$x_{i+1} = \frac{\lambda}{2 \cdot p}(i + 1 + i_o)\sqrt{1 - (\frac{i_o}{(i + i_o + 1)})^2} \tag{22}$$

$$\Delta x_{i+1} = x_{i+1} - x_i = \frac{\lambda}{2 \cdot p} \tag{23}$$

$$\left((i + 1 + i_o)\sqrt{1 - (\frac{i_o}{(i + i_o + 1)})^2} - (i + i_o)\sqrt{1 - (\frac{i_o}{(i + i_o)})^2}\right) \tag{24}$$

$$\Delta\Delta x_{i+1} = \Delta x_{i+1} - \Delta x_i = \tag{25}$$

$$\frac{\lambda}{2 \cdot p}\left((i + 1 + i_o)\sqrt{1 - (\frac{i_o}{(i + i_o + 1)})^2} - 2(i + i_o)\sqrt{1 - (\frac{i_o}{(i + i_o)})^2} + (i + i_o - 1)\sqrt{1 - (\frac{i_o}{(i + i_o - 1)})^2}\right) \tag{26}$$

For small $i < 2000$ values all Δx_i can be computed once. This list of values will be used to search for an index later.

8.5 Limes Relations for Distance Differences Δx_∞

For large i the difference Δx_i can be used to approximate the absolute positions of the x_i values. The relations derived for large i values and limes to ∞ are:

$$(i \gg 1) \qquad \Delta x_{i+1} = x_{i+1} - x_i \approx \frac{\lambda}{2 \cdot p}\sqrt{1 + \frac{H^2}{x_{i+1}^2}} \tag{27}$$

$$\Delta x_\infty = \lim_{x \to \infty} \frac{\lambda}{2 \cdot p}\sqrt{1 + \frac{H^2}{x_{i+1}^2}} = \frac{\lambda}{2 \cdot p} \tag{28}$$

9 Validation of Test Cases

9.1 Generate Signals with known α_i

The measurement over a time period of some seconds consists of an list of values with $i \in \mathbb{N}$ and $N \approx 5000$, N depending on the speed:

- t_i time stamp of measurement
- Δt_i time difference between two crossing points
- A_i amplitude measurement (with sign?)

Each Δt_i belongs to an (un)known index j where x_j is position in the coordinate system fixed to the moving system of the transmitter and Δx_j is the distance between the two positions related to the phase differences on the track ground. Artificial signals with predefined speed and known index j or known position can be used to check the behaviour of an algorithm under the special case of single reflectors and the transition to signals from more then one reflector. Different values for calibration shift are the expected outcome of the test cases.

9.2 Validation of raw Signals from Trials (unknown α_i)

If the correct mapping between Δt_i and Δx_j can be estimated offline, then the reference speed can be validated (if available) or a new reference can be established:

$$\text{Look up table / mapping p[i]=j is known:} \quad v_i = \left(\frac{\Delta x_{p[i]}}{\Delta t_i}\right) \tag{29}$$

$$\text{assumed the angle } \alpha_{p[i]} \text{ is known:} \quad v_i = \left(\frac{\Delta x_{infty}}{\Delta t_i \cdot \cos(\alpha_{p[i]})}\right) \tag{30}$$

The resulting speed should have improved properties:

- no calibration effects
- minimal latency
- low noise

9.3 Histogram of the Amplitude Distribution

If we assume v_{DRS} or a reference speed is known and therefore the VP measurements give reasonable good estimates of Δx_i, then a 2D histogram can be filled by the following process:

$$\Delta x_i = v_{DRS} \cdot \Delta t_i \tag{31}$$

$$idx(A_i) = int(1000 \cdot A_i) \tag{32}$$

$$\text{for all j this contribution is calculated} \quad \Delta H(idx(A_i), j) = exp\left(-\frac{1}{2}\left(\frac{\Delta x_i - \Delta x_j}{\Delta\Delta x_j}\right)^2\right) \tag{33}$$

The important point is, that the contribution to the correct j index is derived by an Gaussian of the difference between $\Delta x_i - \Delta x_j$, no search or inversion is needed. The value of $\Delta\Delta x_j$ is adapted to the actual index, because each bin should get the same contribution from one reflex. The combination of Gaussian and the adaption the denominator gives a histogram, which follow more or less the curve of the reflected energy. The histogram

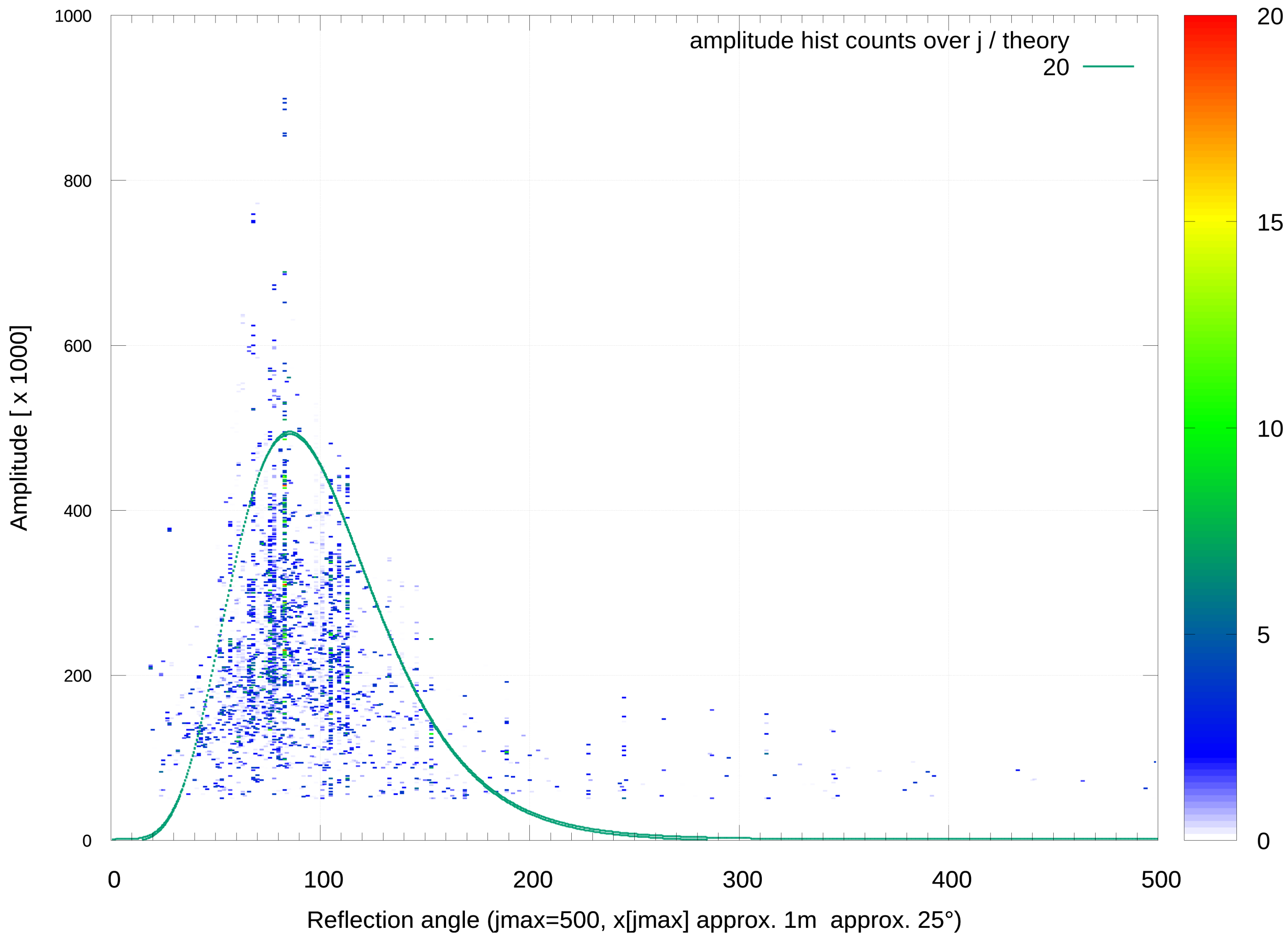

Figure 1: Histogram of the Amplitude Distribution

contains information about the angel distribution and reflection properties of the track ground and is independent from the antenna angle and speed. The mounting height has an influence on the histogram via the Δx_i formula.

The histogram can be at least used to compare trails data between each other. The method don't need much computing power, stored histograms may be a topic for deep learning. These histograms can be also derived for artificial signals as figure of merit.

10 Math Appendix

10.1 Estimation of Δx_j mapping or α_i

The input values are arrays with the following data t_i, Δt_i and A_i. We have outlier with small values of Δt_i and A_i if there is actually no reflector in the field of view. Some derived values maybe helpful to exclude outlier from the estimation:

$$\text{mean values of} \quad \Delta t_i, A_i ... \tag{34}$$
$$\text{differences of time stamps} \quad \delta t_{i+1} = t_{i+1} - t_i \tag{35}$$
$$\text{then we have} \quad \Delta t_i < \delta t_i \tag{36}$$
$$\text{outlier criteria} \quad \Delta t_i > ?? \tag{37}$$
$$\tag{38}$$

10.2 Continuity properties useful for iterative estimation

A list of ideas for a iterative estimation process follows below. It may be useful to start with speed v_{mean} assumed to be known approximately or calculated with methods working without knowing the $j = p[i]$ mapping:

$$\Delta x_{est} = v_{mean} \cdot \Delta t_i \tag{39}$$
$$\Delta x_{est} \approx \Delta x_j => |\Delta x_{est} - \Delta x_j| =!min => p[i] = j \tag{40}$$

The important ideas are all related to the fixed coordinate frame of the moving transceiver. In this frame the distances are independent of time or the actual speed:

$$\text{position is the constant in the moving frame} \qquad x_j = const! \tag{41}$$
$$\text{monotonically increasing distances} \qquad x_{j+1} > x_j \tag{42}$$
$$\text{monotonically decreasing differences} \qquad \Delta x_{j+1} < \Delta x_j \tag{43}$$

Optimization ideas derived from the monotonic properties:

$$\text{small change in } \Delta t_i \text{ lead to small change in } j \qquad \Delta t_i \sim \Delta j \tag{44}$$
$$\text{small change in } v_{est} \text{ lead to small change in j} \qquad \Delta v \sim \Delta j \tag{45}$$
$$\tag{46}$$

Optimization ideas derived from monotonic properties of amplitudes:

$$\text{within single reflex small change of delta t} \qquad \Delta^2 t = \leq ??? \tag{47}$$
$$\text{within single reflex small change of A} \qquad \Delta A = A_{i+1} - A_i \leq ??? \tag{48}$$
$$\text{around maximum two similar } A_i \text{ expected} \qquad \Delta A = A_{i+1} - A_i \leq ??? \tag{49}$$
$$\tag{50}$$

Optimization ideas to detect new reflexes from braking the monotonic properties:

$$\text{large change in } t_i \text{ indicated jump in j} \qquad \Delta t \neq \Delta j \tag{51}$$
$$\text{large change in } A_i \text{ indicated jump} \qquad ??? \tag{52}$$
$$\tag{53}$$

$$\frac{2 \cdot \Delta x_\infty}{\Delta t_i} \leq v_{startvalue} \approx \left\langle \frac{\Delta x_{p(i)}}{\Delta t_i} \right\rangle \geq \frac{\Delta x_\infty}{\Delta t_i} \tag{54}$$
$$\tag{55}$$

The v_{mean} or $2 \cdot v_{mean}$ can be used as a starting values for iterative estimation procedure.

10.3 Relationships based on VP measurements

Using the length relationship for the phase difference between two succeeding measurements allow to derive addition formulas. The moving coordinate frame is used again, to have fixed values for the reflections related to fixed phase conditions. We start with triangle equation for radius vectors and points on the x-axis between two measurements and the relate angle α. The relations are formulate with difference $\Delta\alpha$ of the angle to use trigonometric identities.

$$\text{length difference between radius vectors is known to be}\Delta x_\infty \qquad \text{checked?} \quad \square \tag{56}$$

$$\Delta R = \frac{\lambda}{2p} = \Delta x_\infty \tag{57}$$

$$R_2 => R \qquad \text{checked?} \quad \square \tag{58}$$

$$R_1 => R - \Delta R \qquad \text{checked?} \quad \square \tag{59}$$

$$R_3 => R + \Delta R \qquad \text{checked?} \quad \square \tag{60}$$

$$\Delta x_1 = x_2 - x_1 \qquad \text{checked?} \quad \square \tag{61}$$

$$\Delta x_2 = x_3 - x_2 \qquad \text{checked?} \quad \square \tag{62}$$

$$x_1 = x_2 - \Delta x_1 \qquad \text{checked?} \quad \square \tag{63}$$

$$x_3 = x_2 + \Delta x_2 \qquad \text{checked?} \quad \square \tag{64}$$

$$H^2 + {x_2}^2 = R^2 \qquad \text{checked?} \quad \square \tag{65}$$

$$H^2 + {x_1}^2 = (R - \Delta R)^2 \qquad \text{checked?} \quad \square \tag{66}$$

$$H^2 + {x_3}^2 = (R + \Delta R)^2 \qquad \text{checked?} \quad \square \tag{67}$$

$$H^2 + (x_2 - \Delta x_1)^2 - H^2 - (x_2 + \Delta x_1)^2 = (R - \Delta R)^2 - (R + \Delta R)^2 \qquad \text{checked?} \quad \square \tag{68}$$

$$-2x_2\Delta x_1 + \Delta x_1^2 - 2x_2\Delta x_2 - \Delta x_2^2 = R^2 - 2R\Delta R + \Delta R^2 - R^2 - 2R\Delta R - \Delta R^2 \tag{69}$$

$$-2x_2(\Delta x_1 + \Delta x_2) + \Delta x_1^2 - \Delta x_2^2 = -4R\Delta R \qquad \text{checked?} \quad \square \tag{70}$$

$$x_2(\Delta x_1 + \Delta x_2) = 2R\Delta R - \frac{1}{2}(\Delta x_1 + \Delta x_2)(\Delta x_1 - \Delta x_2) \tag{71}$$

$$x_2(\Delta x_1 + \Delta x_2) = 2R\Delta R + \frac{1}{2}(\Delta x_1 + \Delta x_2)(\Delta x_2 - \Delta x_1) \qquad \text{checked?} \quad \square \tag{72}$$

$$2R\Delta R = x_2(\Delta x_1 + \Delta x_2) - \frac{1}{2}(\Delta x_1 + \Delta x_2)(\Delta x_2 - \Delta x_1) \qquad \text{checked?} \quad \square \tag{73}$$

$$\tag{74}$$

$$R = x_2\frac{\Delta x_1 + \Delta x_2}{2\Delta R} - \frac{1}{4\Delta R}(\Delta x_1 + \Delta x_2)(\Delta x_2 - \Delta x_1) \qquad \text{checked?} \quad \square \tag{75}$$

$$A = \frac{\Delta x_1 + \Delta x_2}{2\Delta R} \tag{76}$$

$$K = \frac{A}{2} \cdot (\Delta x_2 - \Delta x_1) \tag{77}$$

$$R = xA - K \qquad \text{checked?} \quad \square \tag{78}$$

$$x^2 + H^2 = (Ax - K)^2 = A^2x^2 - 2AKx + K^2 \qquad \text{checked?} \quad \square \tag{79}$$

$$x^2(1 - A^2) = A^2x^2 - 2AKx + K^2 - H^2 \qquad \text{checked?} \quad \square \tag{80}$$

$$x^2 + x\frac{2AK}{1 - A^2} - \frac{(K^2 - H^2)}{1 - A^2} = 0 \qquad \text{checked?} \quad \square \tag{81}$$

$$x^2 + x\frac{2AK}{1 - A^2} - \frac{(K^2 - H^2)}{1 - A^2} = 0 \qquad \text{checked?} \quad \square \tag{82}$$

$$\tag{83}$$

Numerical tests with theoretical values gives good results, but formula marked as n.u. in table because of its v dependence!

$$p = \frac{AK}{1-A^2} \qquad \text{checked?} \quad \square \tag{84}$$

$$q = -\frac{K^2-H^2}{1-A^2} \qquad \text{checked?} \quad \square \tag{85}$$

$$x^2 + 2px + q = 0 \qquad \text{checked?} \quad \boxtimes \tag{86}$$

$$x = p \pm \sqrt{p^2-q} \qquad \text{checked?} \quad \boxtimes \tag{87}$$

$$\text{numerical comparison shows that only the negativ sign is correct!!} = \frac{AK}{1-A^2} = \frac{(\frac{\Delta x_1+\Delta x_2}{2\Delta R})^2 \frac{1}{2} \cdot (\Delta x_2 - \Delta x_1)}{1-(\frac{\Delta x_1+\Delta x_2}{2\Delta R})^2} = \tag{88}$$

$$= \frac{1}{2} \cdot \frac{(\Delta x_1+\Delta x_2)^2(\Delta x_2-\Delta x_1)}{(2\Delta R)^2-(\Delta x_1+\Delta x_2)^2} = \tag{89}$$

$$= \frac{AK}{1-A^2} = \frac{(\frac{\Delta x_1+\Delta x_2}{2\Delta R})^2 \frac{1}{2} \cdot (\Delta x_2 - \Delta x_1)}{1-(\frac{\Delta x_1+\Delta x_2}{2\Delta R})^2} = \tag{90}$$

$$= \frac{1}{2} \cdot \frac{(\Delta x_1+\Delta x_2)^2(\Delta x_2-\Delta x_1)}{(2\Delta R)^2-(\Delta x_1+\Delta x_2)^2} = \tag{91}$$

$$K = \frac{A}{2} \cdot (\Delta x_2 - \Delta x_1) \tag{92}$$

$$= \frac{K^2-H^2}{1-A^2} = \frac{0.25(\Delta x_1+\Delta x_2)^2(\Delta x_2-\Delta x_1)^2-(2\Delta R)^2H^2}{(2\Delta R)^2-(\Delta x_1+\Delta x_2)^2} = \tag{93}$$

$$term\ in\ sqrt = (\frac{1}{2} \cdot \frac{(\Delta x_2-\Delta x_1)}{\frac{(2\Delta R)^2}{(\Delta x_1+\Delta x_2)^2}-1})^2 + \frac{0.25(\Delta x_2-\Delta x_1)^2-(\frac{(2\Delta R)^2}{(\Delta x_1+\Delta x_2)^2}H^2}{\frac{(2\Delta R)^2}{(\Delta x_1+\Delta x_2)^2}-1} \tag{94}$$

$$\text{------------------------> ab hier alt} \tag{95}$$

$$cos(\alpha) = \frac{x_{12}}{R} \approx \frac{2\Delta R}{\Delta x_1+\Delta x_2} \tag{96}$$

$$\alpha = acos(\frac{x_{12}}{R}) \approx acos(\frac{2\Delta x_\infty}{\Delta x_1+\Delta x_2}) \tag{97}$$

$$\tag{98}$$

Note for proof of concepts:
The formulas based on distance difference can be always understood as measurement with $\Delta x_i = v \cdot \Delta t_i$. If v is assumed to be constant then all quotient formulas with differences can be seen as equations for Δt_i, because v cancels out. For numerical validation of the formulas are based on the theoretical Δx_i values, which can be calculated according to the above formulas.

10.4 i from $t_i, \Delta t_i, A_i$ Measurements

$$i_o = 2 \cdot p \cdot \frac{H}{\lambda} => \frac{\lambda}{2 \cdot p} = \frac{H}{i_o} = \Delta x_\infty \tag{99}$$

$$x_i = (i + i_o) \cdot \frac{\lambda}{2 \cdot p} \sqrt{1 - (\frac{i_o}{i + i_o})^2} = H \sqrt{\frac{(i + i_o)^2}{i_o^2} - 1} = \tag{100}$$

$$H \sqrt{\frac{i^2 + 2i \cdot i_o}{i_o^2}} = \tag{101}$$

$$\frac{H}{i_o} \sqrt{i^2 + 2i \cdot i_o} = \ \Delta x_\infty \sqrt{i^2 + 2i \cdot i_o} \tag{102}$$

$$x_i = \Delta x_\infty \sqrt{i^2 + 2i \cdot i_o} \tag{103}$$

$$x_i = \Delta x_\infty \cdot i \cdot \sqrt{1 + 2\frac{i_o}{i}} \quad (i > 0) \quad x_0 = 0 \tag{104}$$

$$(x_{i+1} - x_i)(x_{i+1} + x_i) = x_{i+1}^2 - x_i^2 = \Delta x_\infty^2 \cdot \Big((i+1)^2 + 2(i+1) \cdot i_o - i^2 + 2i \cdot i_o\Big) = \tag{105}$$

$$= \Delta x_\infty^2 \cdot \Big((i^2 + 2i + 1 + 2ii_o + 2i_o) - (i^2 + 2ii_o)\Big) = \Delta x_\infty^2 (2(i + i_o) + 1) \tag{106}$$

$$\frac{(x_{i+1} - x_i)(x_{i+1} + x_i)}{(\Delta x_\infty)^2} = 2(i + i_o) + 1 \tag{107}$$

$$\frac{(x_{i+1} - x_i)(x_{i+1} + x_i)}{(\Delta x_\infty)^2} - 2i_o = 2i + 1 \tag{108}$$

$$i = \frac{1}{2} \left(\frac{(x_{i+1} - x_i)(x_{i+1} + x_i)}{(\Delta x_\infty)^2} - 2i_o - 1 \right) \tag{109}$$

$$i = \frac{1}{2} \left(\frac{(x_{i+1} - x_i)(x_{i+1} + x_i)}{(\Delta x_\infty)^2} - 2i_o - 1 \right) \tag{110}$$

$$\tag{111}$$

10.5 Derivations of angle relationships including inverse of cos-factor ($1/cos(\alpha)$)

Following the ideas from Riccardo Matera, formulas for the relation between two measurements can be derived by substitution of angles to a mean value $+/-\Delta\alpha$ with $+/-\Delta\alpha$ for the two measurements. Then the typical concept of adding and subtracting the equations allows to separate the variables and find a solution. For angle relationship between two measurement referring to the same reflex, a relation For fixed VP differences the derivation formulas can't be used, but as a limes for large index values the formulas are relevant.

$$\text{measurement i=1,2} \quad \Delta x_i = v_{DRS} \cdot \Delta t_i \tag{112}$$

$$\text{use formula with angle} \quad \Delta x_i = \Delta x_\infty \frac{1}{cos(\alpha_i)} \tag{113}$$

$$\text{substitute mean angle} \quad \frac{\Delta x_\infty}{\Delta x_{1,2}} = cos(\alpha \pm \Delta\alpha) \tag{114}$$

$$\frac{\Delta x_\infty}{\Delta x_{1,2}} = cos(\alpha)cos(\Delta\alpha) \mp sin(\alpha)sin(\Delta\alpha) \tag{115}$$

$$\text{add both, last summand cancels} \quad \frac{\Delta x_\infty}{\Delta t_1} + \frac{\Delta x_\infty}{\Delta t_2} = 2cos(\alpha)cos(\Delta\alpha) \tag{116}$$

$$\text{sub both, first summand cancels} \quad \frac{\Delta x_\infty}{\Delta t_1} - \frac{\Delta x_\infty}{\Delta t_2} = -2sin(\alpha)sin(\Delta\alpha) \tag{117}$$

$$\text{divide both equations} \quad \frac{\frac{\Delta x_\infty}{\Delta x_1} - \frac{\Delta x_\infty}{\Delta x_2}}{\frac{\Delta x_\infty}{\Delta x_1} + \frac{\Delta x_\infty}{\Delta x_2}} = -tan(\alpha)tan(\Delta\alpha) \tag{118}$$

$$\frac{\frac{\Delta x_\infty \Delta x_2 - \Delta x_\infty \Delta x_1}{\Delta x_1 \Delta x_2}}{\frac{\Delta x_\infty \Delta x_2 + \Delta x_\infty \Delta x_1}{\Delta x_1 \Delta x_2}} \tag{119}$$

$$\frac{\Delta x_\infty \Delta x_2 - \Delta x_\infty \Delta x_1}{\Delta x_\infty \Delta x_2 + \Delta x_\infty \Delta x_1} = \tag{120}$$

$$\frac{\Delta x_1 - \Delta x_2}{\Delta x_1 + \Delta x_2} = -tan(\alpha)tan(\Delta\alpha) \tag{121}$$

$$\frac{\Delta x_2 - \Delta x_1}{\Delta x_2} = 2tan(\alpha)sin(\Delta\alpha)\frac{cos(\alpha)}{cos(\alpha+\Delta\alpha)} \tag{122}$$

$$\frac{\Delta x_1 - \Delta x_2}{\Delta x_1 + \Delta x_2} = -tan(\alpha)tan(\Delta\alpha) \tag{123}$$

$$\frac{(\Delta x_1 - \Delta x_2)x_2}{(\Delta x_1 - \Delta x_2)(\Delta x_1 + \Delta x_2)} = \frac{-tan(\alpha)tan(\Delta\alpha)}{2tan(\alpha)sin(\Delta\alpha)}\frac{cos(\alpha+\Delta\alpha)}{cos(\alpha)} = \tag{124}$$

$$-\frac{1}{2}cos(\Delta\alpha)\frac{cos(\alpha)cos(\Delta\alpha) - sin(\alpha)sin(\Delta\alpha))}{cos(\alpha)} = \tag{125}$$

$$-\frac{1}{2}cos^2(\Delta\alpha)\left(1 - \frac{sin(\alpha)sin(\Delta\alpha))}{cos(\alpha)}\right) = \tag{126}$$

$$\frac{t_2}{(\Delta x_1 + \Delta x_2)} \approx \frac{1}{2}cos^2(\Delta\alpha) \tag{127}$$

$$cos(\Delta\alpha) \approx \sqrt{\frac{2\Delta x_2}{\Delta x_1 + \Delta x_2}} \tag{128}$$

$$1 - cos^2(\Delta\alpha) = 1 - \frac{2\Delta x_2}{(\Delta x_1 + \Delta x_2)} \tag{129}$$

$$sin(\Delta\alpha) = \sqrt{\frac{\Delta x_1 - \Delta x_2}{\Delta x_1 + \Delta x_2}} \tag{130}$$

$$tan(\Delta\alpha) = \frac{sin(\Delta\alpha)}{cos(\Delta\alpha)} = \frac{\sqrt{\frac{\Delta x_1 - \Delta x_2}{\Delta x_1 + \Delta x_2}}}{\sqrt{2\frac{\Delta x_2}{\Delta x_1 + \Delta x_2}}} \tag{131}$$

$$tan(\Delta\alpha) = \sqrt{\frac{(\Delta x_1 - \Delta x_2)^2(\Delta x_1 + \Delta x_2)(2\Delta x_2)}{(\Delta x_1 + \Delta x_2)^2(\Delta x_1 + \Delta x_2)(\Delta x_1 - \Delta x_2)}} \tag{132}$$

$$tan(\Delta\alpha) = \frac{\sqrt{(\Delta x_1 - \Delta x_2)(2\Delta x_2)}}{(\Delta x_1 + \Delta x_2)} \tag{133}$$

$$cos^2(\Delta\alpha) + sin^2(\Delta\alpha) = 1 => 1 + tan^2(\Delta\alpha) = \frac{1}{\cos^2(\Delta\alpha)} \tag{134}$$

$$tan(\Delta\alpha) = \frac{\sqrt{(\Delta x_1 - \Delta x_2)(2\Delta x_2)}}{(\Delta x_1 + \Delta x_2)} \tag{135}$$

$$tan^2(\Delta\alpha) = \frac{(\Delta x_1 - \Delta x_2)(2\Delta x_2)}{(\Delta x_1 + \Delta x_2)^2} \tag{136}$$

$$1 + tan^2(\Delta\alpha) = \frac{\Delta x_1^2 + 4\Delta x_1\Delta x_2 - \Delta x_2^2}{(\Delta x_1 + \Delta x_2)^2} \tag{137}$$

$$cos^2(\Delta\alpha) = \frac{1}{1 + tan^2(\Delta\alpha)} = \frac{(\Delta x_1 + \Delta x_2)^2}{\Delta x_1^2 + 4\Delta x_1\Delta x_2 - \Delta x_2^2} \tag{138}$$

$$cos(\Delta\alpha) = \frac{1}{1 + tan^2(\Delta\alpha)} = \frac{(\Delta x_1 + \Delta x_2)}{\sqrt{\Delta x_1^2 + 4\Delta x_1\Delta x_2 - \Delta x_2^2}} \tag{139}$$

$$\tag{140}$$

10.5.1 Derivation over time of calibration factor

$$tan(\alpha_i) = \frac{H}{x_i} \tag{141}$$

$$\Delta x_i \approx \frac{\lambda}{2 \cdot p}\sqrt{1 + \frac{H^2}{x_i^2}} \tag{142}$$

$$\frac{1}{cos(\alpha_i)} = \sqrt{1 + \frac{H^2}{x_i^2}} \tag{143}$$

$$\Delta x_i \approx \frac{\lambda}{2 \cdot p}\frac{1}{cos(\alpha_i)} \tag{144}$$

For x_i, α_i without restrictions to special points with defined phase we have:

$$tan(\alpha) = \frac{x}{H} = \sqrt{\frac{1}{cos^2(\alpha)} - 1} = \sqrt{\frac{1 - cos^2(\alpha)}{cos^2(\alpha)}} \tag{145}$$

$$x_i = \frac{H}{tan(\alpha_i)} = H\frac{cos(\alpha_i)}{\sqrt{1 - cos^2(\alpha_i)}} \tag{146}$$

$$\text{time and position of first appearance of the reflex} \quad x(t) = x_o - v \cdot (t - t_o) \tag{147}$$

$$\text{angle definition as function of x(t)} \quad \alpha(x(t)) \tag{148}$$

$$\text{angle and coordinates} \quad \alpha(x(t)) = arctan(\frac{H}{x(t)}) \tag{149}$$

$$\text{partial derivation and chain rule} \quad \frac{\delta\alpha(x(t)))}{\delta t} = \frac{1}{1 + \frac{H}{x(t)}} \cdot \frac{\delta}{\delta t}(\frac{H}{x(t)}) \tag{150}$$

$$\text{partial derivation and chain rule} \quad \frac{\delta}{\delta t}(\frac{H}{x(t)}) = \frac{-H}{x^2(t)} \cdot v \tag{151}$$

$$\text{calibration factor} \quad k(\alpha(x(t))) = \frac{x_\infty}{cos(\alpha(x(t)))} \tag{152}$$

$$\text{derivation over t, apply n x chain rule} \tag{153}$$

$$\frac{\delta}{\delta t}(k(\alpha(x(t)))) = \frac{x_\infty}{cos(\alpha(x(t)))} \cdot sin(\alpha(x(t))) \cdot \frac{\delta\alpha(x(t)))}{\delta t} \tag{154}$$

$$\frac{\delta}{\delta t}(k(\alpha(x(t)))) = \frac{x_\infty}{cos(\alpha(x(t)))} \cdot sin(\alpha(x(t))) \cdot \frac{-H}{x^2(t)} \cdot v \tag{155}$$

$$\frac{\delta}{\delta t}(k(\alpha(x(t)))) = \frac{-x_\infty}{H} tan(\alpha(x(t)))^3 \cdot v \tag{156}$$

$$\frac{\delta}{\delta t}(k(\alpha(x(t)))) = \frac{-x_\infty}{H}(\frac{H}{x(t)})^3 \cdot v \tag{157}$$

$$\frac{\delta}{\delta t}(k(\alpha(x(t)))) = \frac{-x_\infty}{H} tan^3(\alpha(x(t)) \cdot v \tag{158}$$

11 Reflexes from Amplitude Search

11.1 Available Data for Amplitude Search

Typical data format is:

1. time in msec t_i
2. VP measurement in msec Δt_i
3. |Amplitude| in a.u. A_i

Typical, only absolute amplitude not the sign values are stored during the measurement. The time between the measurements can be longer then the sum of the VP measurements (existence of gaps), if the signal is low in between. With sign values the sign must alternate, which would be a good indicator the consecutively measurements. There are several types of search conditions:

- Δt_i above threshold: $\Delta t_i \geq \Delta t_{thr}$
- A_i above threshold: $A_i \geq A_{thr}$
- 2 x A_i values near maximal value but below maximal value
  $A_i \approx \frac{1}{\sqrt{2}} A_{max-neari}$
- VP measurement Δt_i
- Amplitude $|A_i| \leq |A_{i+1|}$

Some conditions can also be used to get a stochastic information:

- A_i ordering:
  $A_i \leq A_{i+1}$ => right side of $Ba(x)$ function and
  $A_i \geq A_{i+1}$ => left side of $Ba(x)$ function
- Length of ordered chain in t_i:
  While the reflector remains inside the Ba(x) envelope, all Δt_i must be ordered according the Δx_i formula to achieve a constant speed $\Delta x_i = v \Delta t_i$. The ordering with i counting for $x = 0$ is fixed to $\Delta x_{i+1} \leq \Delta x_{i+1}$, the Δt_i ordering depends to DOT (direction of travel), if the reflector is moving towards transceiver, then $\Delta t(t_{j+1}) \geq \Delta t(t_j)$.

$$\frac{\Delta x_1 \cdot \Delta x_2}{(\Delta x_1 + \Delta x_2)^2} \tag{159}$$

$$\lim_{\Delta x_1 \to \Delta x_1} \frac{\Delta x_1 \cdot \Delta x_2}{(\Delta x_1 + \Delta x_2)^2} - \frac{1}{4} = 0 \tag{160}$$

$$\frac{4\Delta x_1 \cdot \Delta x_2 - (\Delta x_1 + \Delta x_2)^2}{4\ (\Delta x_1 + \Delta x_2)^2} = \tag{161}$$

$$\frac{4\Delta x_1 \cdot \Delta x_2 - \Delta x_1^2 - 2\Delta x_1 \Delta x_2 - \Delta x_2^2}{4(\Delta x_1 + \Delta x_2)^2} = \tag{162}$$

$$-\frac{\Delta x_1^2 - 2\Delta x_1 \Delta x_2 + \Delta x_2^2}{4(\Delta x_1 + \Delta x_2)^2} = \tag{163}$$

$$\frac{(\Delta x_1 - \Delta x_2)^2}{4(\Delta x_1 + \Delta x_2)^2} = \tag{164}$$

$$\frac{(\Delta x_1 - \Delta x_2)}{2(\Delta x_1 + \Delta x_2)} = \tag{165}$$

R-No.	Relation	Result	Formula	Properties[0]
1	single value	$cos(\alpha)$	$\frac{\Delta x_\infty}{\Delta x_i}$	n.u.[1]
2	ΔR differences	$cos(\alpha)$	$\frac{2\Delta x_\infty}{\Delta x_1+\Delta x_2}$	n.u.[2]
3	$\frac{\Delta x_1}{\Delta x_2}$ quotient	$2\ tan(\alpha)\ sin(\Delta\alpha)\frac{cos(\alpha)}{cos(\alpha+\Delta\alpha)}$	$\frac{\Delta x_2-\Delta x_1}{\Delta x_2}$	n.p. [3]
4	differences	$tan(\alpha)\ tan(\Delta\alpha)$	$\frac{\Delta x_1-\Delta x_2}{\Delta x_1+\Delta x_2}$	n.p. [4]
5	quotient 3/4	$cos(\Delta\alpha) \approx$	$\frac{(\Delta x_1+\Delta x_2)}{\sqrt{\Delta x_1^2+4\Delta x_1\Delta x_2-\Delta x_2^2}}$	o.k. [5]
6	quadratic equation	$x = x_2$ from H,Δx_1, Δx_2	$x^2 + x\frac{2AK}{1-A^2} - \frac{(K^2-H^2)}{1-A^2} = 0$	n.u.[6]

[0] n.u. not useful, o.k. okay, n.p numerical problems
[1] can't be to reduced from x_∞ to Δt, estimated speed is needed
[2] can't be to reduced from x_∞ to Δt, estimated speed is needed
[3] can be reduced to Δt formula, $sin(\Delta\alpha) \ll 1$
[4] can be reduced to Δt formula, $tan(\Delta\alpha) \ll 1$
[5] can be reduced to Δt formula, only approx valid
[6] numerically tested, but no Δx quotient, therefore dependency on v

Figure 2: first list of algorithms

12 Amplitude Histogram

13 Continuous Measurement based on periods of n $\Delta t_i, i \in [0..n-1]$

14 Discover Relationships the old way (old school)

14.1 Plotting a straight line

The relationships investigated should pbe plotted in a way to achieve straight line. For example a straight line Don't forget to use the "Definition of Continuity" !! y axis $frac1cos^2()$

14.2 Possible Inter-Beam Correlation in $t_i, \Delta t_i, A_i$ Measurements

14.2.1 Correlation between High-Beam and Low-Beam

The x_i are not depending on the angle of the microwave transceiver, but only on the actual angle and the mounting height. If two transceivers are mounted in position as the beam comes from the same point source, then the x_i positions are the same for both beams. If the positions are different, the height is different and therefore different $i_{k,o}$ are defined. The measurement data from different beam angle will be labeled as:

$$\text{index k 0 = h-beam, 1 = low beam} \qquad t_{k,i}, \Delta t_{k,i}, A_{k,i} \tag{166}$$

$$\text{if different mounting height k 0 = h-beam, 1 = low beam} x_{k,i} = (i + i_{k,o}) \cdot \frac{\lambda}{2 \cdot p}\sqrt{1 - (\frac{i_{k,o}}{i + i_{k,o}})^2} \tag{167}$$

$$\sum A_{0,i} \cdot A_{1,i+\Delta C_{n,m}} \tag{168}$$

$$\Delta C_{n,m} = x_{0,n} - x_{1,m} \tag{169}$$

15 Real Time Data Processing Constraints

15.1 Reflex Identifcation to correct Calibration Shift

Each reflex is handled with its own reflex parameter like reflection-angle, cross-section and more. The signal of the reflex is a contribution to the complete signal. This additive effects have direct influence on the VP measurement and therefore subtractive process seem to be a possible choice, but this is not confirmed yet. The structure of the data handling is like vector of n dimensions for n reflexes, the number should larger then the normal number of reflexes which can be identified. If no reflex is assigned, the these dimension parameter are set to zero.

15.2 Reaction time to detect calibration shift

The information about track ground is a statistical information. Therefore to detect a change will take some time or approx. 1000-5000 samples, means $1000 - 5000 \cdot dx_\infty = 1000\frac{\lambda}{2p} \approx$ several meters $\leq 2 - 5$. One important point is the limited change rate guaranteed for the product, because fast jumps in the speed will have an effect on the customer system (plausibility check will trigger).

Requirement: Allow instant change of the speed for compensation of calibration shift. Need protocol, which supports message about changes in calibration

16 Inverse Functions of Angel or Index (Numerical Inversion)

For the monotonic function based on a quotient term of Δx_i the relation between angle or index can be invertedt numerical by an lookup table.

Due to the fact that the changs of Δx_i decreases for larger index values, it may be necessary to transform the function with log and scale accordingly. For example the function:

$$u_i = \frac{0.5 \cdot (\Delta x_i - \Delta x_{i-1})}{(\Delta x_i + \Delta x_{i-1})} \tag{170}$$

$$\text{log(u) can be scaled as} \quad uidx = int(-250.0(log(u))) \tag{171}$$

$$ufo[int(250.0(log(u_i)))] = i \tag{172}$$

to give an different integer value for at least each Δx_j. For an easy look-up, an inverted index can be stored (filled with the last values, if no value in the loop was supplied). Values of angle or Δx_i are also available as arrays, because all these values are not dependent on the speed. The look-up error tested with theoretical data shows about ± 1 of the index value.

Actually this formula don't work with trial data

17 Idee vom 13.06.2018

For each measurements Δt_i derived from a single reflex we can find a Δx_j with:

$$i \in [1..k] \ \exists \ j \in [1..n] \qquad \Delta x_j = v \cdot \Delta t_i \tag{173}$$
$$\tag{174}$$

A matrix $p_{i,j}$ which contains rows of zeros with only one element set to 1 defines the mapping between measurement i with Δt_i and the angle or position of the reflex at index j. The mapping $pmap[i] = j$ can be stored as an array. The list of values in vector $\Delta x_1, \dots, \Delta x_n$ is fixed and can be calculated in advance. The number n depends on the antenna characteritics and can be truncated/limited to $n \ll 2000$.

$$\begin{bmatrix} p_{1,1} & p_{1,2} & \dots & p_{1,n-1} & p_{1,n} \\ \dots \\ \dots \\ p_{k,1} & p_{k,2} & \dots & p_{k,n-1} & p_{k,n} \end{bmatrix} \cdot \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \dots \\ \Delta x_{n-1} \\ \Delta x_n \end{bmatrix} = \begin{bmatrix} v_1 \cdot \Delta t_1 \\ \dots \\ \dots \\ v_k \cdot \Delta t_k \end{bmatrix} \approx v \cdot \begin{bmatrix} \Delta t_1 \\ \dots \\ \dots \\ \Delta t_k \end{bmatrix} \tag{175}$$

A relexed contraints for $p_{i,l} = \delta(l-j)$ can formulated as:

$$\sum_{i=1}^{i=n} p_{k,i} = 1 \tag{176}$$

An estimation of v can be used to estimate p values:

$$p_{i,k} = a \cdot e^{-\left(\frac{\Delta x_i - \Delta x_k}{\Delta x_{k-1} - \Delta x_k}\right)^2} \tag{177}$$

$$\mathbb{I}_k = \begin{bmatrix} 1_1 & 1_2 & \dots & 1_{k-1} & 1_k \end{bmatrix} \tag{178}$$

$$\mathbb{I}_k \cdot \begin{bmatrix} p_{1,1} & p_{1,2} & \dots & p_{1,n-1} & p_{1,n} \\ \dots \\ \dots \\ p_{k,1} & p_{k,2} & \dots & p_{k,n-1} & p_{k,n} \end{bmatrix} \cdot \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \dots \\ \Delta x_{n-1} \\ \Delta x_n \end{bmatrix} = \mathbb{I}_k \cdot \begin{bmatrix} v_1 \cdot \Delta t_1 \\ \dots \\ \dots \\ v_k \cdot \Delta t_k \end{bmatrix} = s \tag{179}$$

$$x_\lambda = \begin{cases} x & \text{if } \lambda \text{ is an eigenvalue;} \\ -x & \text{if } -\lambda \text{ is an eigenvalue;} \\ 0 & \text{otherwise.} \end{cases} \tag{180}$$

Example with speed estimation used to convert VP measurmenets to distances and angles for validation of track types. The example below should show an homogeneous distribution over the angle, but results shows only some dominant reflexes.

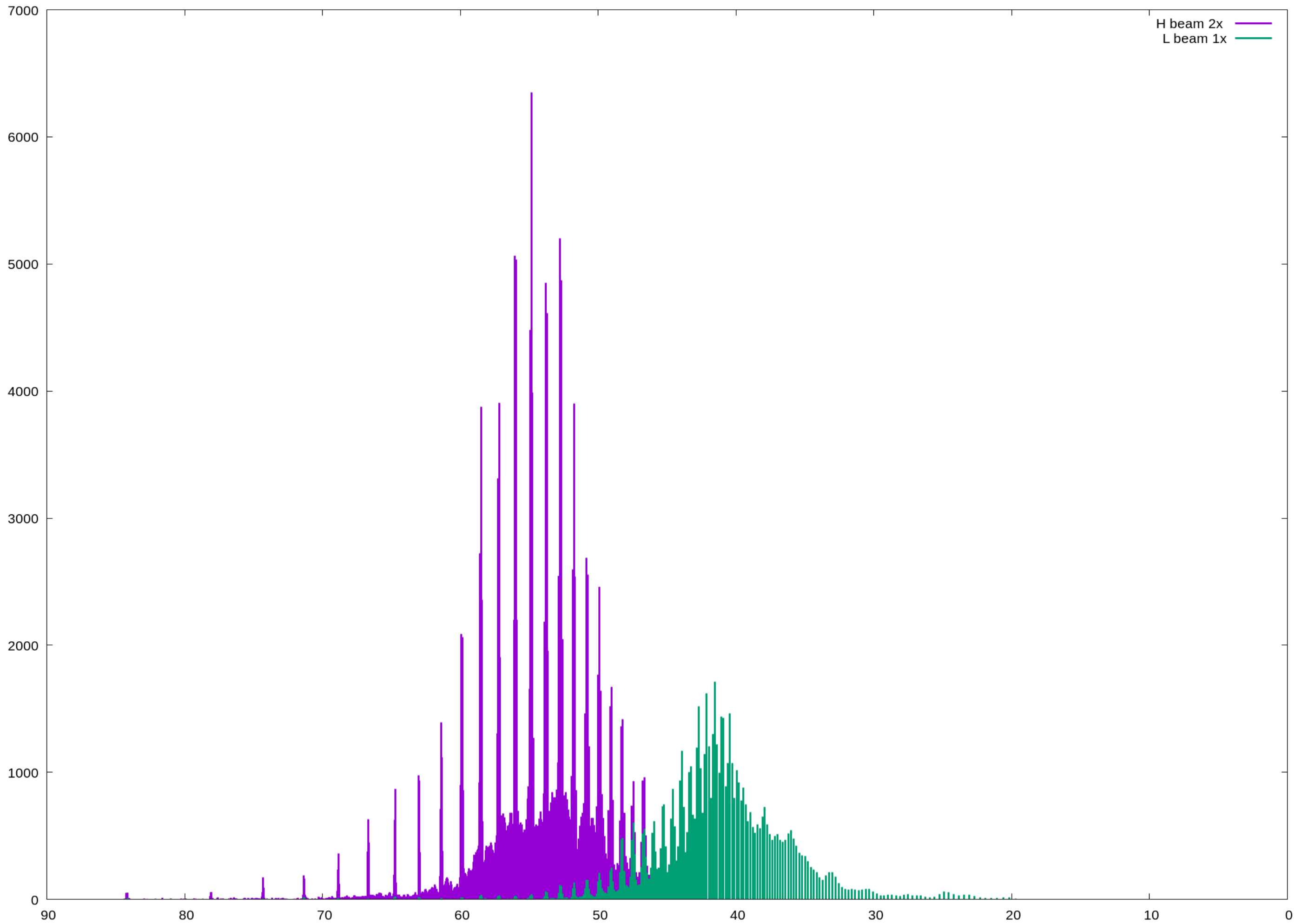

Figure 3: Histogram of the Amplitude Distribution

18 Idee vom 20.06.2018

See publication about NDFFT or others: Definition of the axis and understanding of the meaning is enhanced by these publications. The time (distance from origin) in the origin of the coordinate system Measurement of amplitude $A(t)$ at t_n The dual space can be defined in several ways: FFT will map to inverse of time <> frequency! The duration of a period is not mapped to frequency. FFT inhernet assumption are periodic signal and constant phase.

The FFT adds all measurements with a sign factor according to the actual frequency, if the measurements are correlated to this raster of time, then the sum is large. If their is a phase change, then the sum may be zero or small even at the right frequency.

1. measurement is t_i, A_i and Δt_i, the period duration.
2. phase differences between measurements is fixed by Δx_∞.
3. phase change is related to the projection of the view angle on x-axis.
4. for constant speed the distance on x-axis is $\Delta x_j = v \cdot \Delta t$.
5. a reference value for Δx_j could be Δx_∞, it makes sense to build $\frac{\Delta x_j}{\Delta x_\infty}$ which is the inverse of the $cos(\alpha)$.
6. a reference value could also be the inverse of the $cos(\alpha)$ or $cos(\alpha)$ itself displayed starting at $x = 0$ with $90 gradandendingatvalueabove 10 grad$.

If all measurements are aequidistant $\Delta t = const$ For each measurements Δt_i derived from a single reflex we can find a Δx_j with:

19 Ideen vom 21.06.2018

19.1 Other Math Tricks

Fresnel transformation `https://www.osapublishing.org/josaa/abstract.cfm?URI=josaa-31-4-755`
Chebyshev polynomials of the first kind `https://en.wikipedia.org/wiki/Chebyshev_polynomials`

VPH for test case validation:

1. optimize for values of H
2. optimize for values of v (constant value)
3. optimize for periodic deviation of v (not a noise value, but peridoic)
4. use 3 reflex values for v, angle (instead of 2)

Optimize implementation of 1/0 matrix operations for 0/1 matrix coded as bits:

1. Using 64,32,16,8 bit look ahead for block out, because ifnull operation very fast.

DRS4/1 optimization (how to validate that antenna characteristics is sub-optimal):

1. high y- values of reflex have the same effect as different height
2. measurement with slotted absorber to validate above

Actual price IVS234 is 92 €!